

About carbon

Carbon is an amazing atom. It loves to bond with other atoms, which is why it has been called 'the glue of life'. Carbon is the fourth most abundant element in the universe. It is found in all organic matter, and makes up a large proportion of all living material. Diamonds and graphite are both examples of pure carbon.

The carbon cycle

The carbon cycle includes the movement of carbon between the soil, the things that live on it and the atmosphere.

Plants — trees, shrubs, grasses, crops and other vegetation — take carbon dioxide from the air through their leaves and use sunlight — the photosynthesis process — to transform carbon dioxide and water into oxygen, and into the sugars and starches the plants use to grow. Plants store carbon in their wood, leaves and roots.

Plants shed organic matter as they decay and die, which microorganisms (like bacteria and fungi) break down. This releases the plants' carbon back into the soil, increasing soil organic carbon and releasing carbon dioxide into the atmosphere.

Animals similarly contribute to the carbon cycle. When they breathe, they release carbon dioxide; when they die in the natural environment, their decaying remains release carbon back into the soil.

Oceans also absorb carbon dioxide from the atmosphere. Rocks and other geological deposits (such as coal) store carbon. Burning coal and other fossil fuels releases the stored carbon into the atmosphere.

Watch this video

In this video (4:15 minutes), Professor Richard Eckard of the University of Melbourne discusses how the carbon cycle works in agriculture.

PROFESSOR RICHARD ECKARD: This is an explanation of the carbon cycle in agriculture.

It's based on a grazing system, but the same principles apply to a cropping system.

It all starts with sunlight energy and the process of photosynthesis in plants, which allows plants to capture carbon dioxide from the atmosphere and lock it away into a plant.

You will notice that we've highlighted the letter c in its various forms of carbon, showing that carbon in CO₂ is a gas, but carbon in the plant is CHO, which is a sugar, not a gas. The key point is that carbon takes many forms as it cycles through our agricultural systems.

In a cropping system, we would then harvest that plant. In a grazing system, an animal will graze the pasture and convert the carbohydrate into animal products.

The majority of the carbon consumed by the animal is belched out or breathed out as CO₂ back to the atmosphere.

The majority of the animal products or crop products, eaten by humans is also respired back to the atmosphere within 12 months as CO₂.

If that was all that happened, the carbon cycle would be completely balanced, with all inputs returning to the atmosphere in the same form within a short period of time.

However, a small amount of carbon entering the rumen of the animal is converted into methane, CH₄, which is carbon in another gaseous form. Methane is fundamentally different to CO₂ and has a far higher warming potential in the atmosphere for the duration that it's there.

The animal defecates, and so there's faecal carbon in an organic form going back to the soil.

Plants will leave litter on the soil surface as a form of organic carbon.

As the plant grows, it produces roots, and these roots leave carbon behind either as root fragments or as root exudates of sugars.

This organic form of carbon in the soil can either be in a larger fraction, which we call the particulate organic carbon, and turns over fairly fast through microbial action, perhaps in hours through to a few years.

The smaller, more resistant fractions of organic carbon in the soil are called humus.

These are the more resistant fractions that last for decades through to hundreds of years.

In the soil, there are billions of microbes that then work actively on this organic carbon as their food source, actively breaking down the carbon to release the stored nutrients back to the soil to allow the crop or pasture to grow.

In the process, these microbes then release the carbon stored in the organic material as carbon dioxide back to the atmosphere, and the carbon cycle completes itself.

In a cropping system, this process of mineralization may release 30 to 50 kilograms of nitrogen per hectare per year out of the soil organic matter. In a dairy pasture system, it can be as much as 250 kilograms of nitrogen per hectare per year from the organic matter.

The point of carbon farming is to capture and make that cycle as efficient as possible so that we can transfer as much of the carbon from the atmosphere into product that we're interested in producing, which could be meat, wool, milk, grain, and crops.

Carbon farming is therefore focused on maximising the carbon from photosynthesis through to a product that goes to market as efficiently as possible.

In the process, good carbon farming would aim to leave as much carbon in the soil as possible, capture some of the carbon in trees growing in the landscape, and minimise the amount of carbon lost as methane and as a nitrogen gas.

About soil organic carbon

To increase carbon storage in soil, you need to increase carbon content in the soil, reduce carbon lost from the soil, and preferably both.

Along with air, water and inorganic matter, soils naturally contain a small proportion of soil organic matter. Soil organic matter is all the living and dead organic material — plants, soil organisms and animal materials — in various stages of decomposition, but not the fresh, undecomposed organic material on the surface. Soil organic matter:

- binds soils together, increasing their resistance to erosion
- provides nutrients for vegetation
- supports soil microbiology and cycling of nutrients (reducing the need for fertilisers) and improves water infiltration and water-holding capacity and the soil's ability to retain nutrients.

Soil organic carbon (SOC) makes up a large component of soil organic matter. The amount of SOC can vary depending on soil type, environmental conditions (including rainfall) and land management practices. It is generally high in clay soils under pasture, and it can be the highest in peat soils. Soil organic matter and SOC are usually expressed as a percentage of the soil by weight. Carbon farming activities that can increase SOC include cover cropping, no-till farming and agroforestry. Carbon farming activities that can reduce carbon loss include avoiding land management practices like over-grazing, over-tilling and stubble burning.

Source: <https://www.agriculture.gov.au/agriculture-land/farm-food-drought/climatechange/carbon-farming-outreach-program/training-package/topic-1/6-increasing-carbon-storage>